

Md Omar Faruqe RIYAD
Software Engineer | AI Engineer

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EDUCATION

2018–2023 **B.Sc. in Computer Science and Engineering**
Shahjalal University of Science and Technology (SUST)
CGPA : 3.53/4.00 (Last two years : 3.83)

SKILLS

Languages	Java, C/C++, Python, JavaScript, SQL, Shell
Frameworks & Libraries	JEE (JSF, CDI, EJB, JPA), FastApi, PrimeFaces, GlassFish, Flask, React
Databases & Data Storage	PostgreSQL, MySQL, Neo4j (Graph), MinIO (S3-compatible)
DevOps & Observability	Docker, Git, GitHub Actions CI/CD, Ansible, Prometheus, Grafana, Loki
AI & ML	PyTorch, LangChain, RAG, LLM, Vector Database (ChromaDB), OpenAI & Ollama APIs
Security	Jakarta Security, Foundational Cryptography
Tools	Flyway, JasperReports, L ^A T _E X

JOB EXPERIENCE

Present Oct 2025	AI Engineer, ALTRI.AI, US (part-time) > Architected a Hybrid AI Scoring Engine : Designed a dual-model system integrating XGBoost for price forecasting and OpenAI LLMs for semantic analysis of property descriptions, synthesizing structured and unstructured data into actionable investment grades. > Performance Engineering : Refactored the data enrichment pipeline using Python asyncio and co-routines, achieving a 10x increase in throughput and reducing cloud compute costs by parallelizing I/O-bound LLM operations. > Event-Driven Data Pipeline : Built a fault-tolerant ingestion engine using HomeHarvest and PostgreSQL. Implemented a "Delta Sync" strategy with immutable event logging to track complex property lifecycles (Active → Sold) without data loss. > Production MLOps Workflow : Established a containerized training pipeline with Human-in-the-Loop validation gates, ensuring new "Challenger" models statistically outperform production "Champions" before deployment. > Secure Full-Stack Architecture : Developed a FastAPI backend and React frontend backed by Supabase. Implemented robust security via Row Level Security (RLS) and distributed job locking for safe, multi-tenant concurrency. > DevOps & Infrastructure : Dockerized the complete microservices stack (API, Scheduler, Trainer) for reproducible deployments across local environments and Render cloud infrastructure. Python RAG LLM FastApi React PostgreSQL
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Present
April 2023

Software Engineer, DYNAMIC SOLUTION INNOVATORS, Bangladesh

- > **Enterprise Digital Transformation** : Engineered security-critical government applications, transforming complex offline bureaucracies into streamlined, automated digital workflows.
- > **Security Engineering** : Conducted vulnerability assessments and security audits as a key member of the internal cybersecurity team, ensuring compliance with government data protection standards.
- > **DevOps Optimization** : Accelerated team-wide development velocity by resolving a critical GlassFish hot-swap failure, slashing redeployment latency by 96% (2m → 5s).

Key Projects :

E-Appeal (NBR • USAID)

- > Pioneered the company's first large-scale JSF application, establishing engineering guidelines.
- > Designed an **event-driven architecture** using JMS to decouple services, ensuring high availability and fault tolerance.

Project Scratch

Internal reusable Core Services Platform to accelerate development.

- > **Observability & Infrastructure** : Engineered a centralized monitoring stack (Prometheus, Loki, Grafana) and dockerized the full legacy stack (GlassFish, PostgreSQL) for consistent local development.
- > Developed a reusable Lazy Data Model component and **contributed a related feature enhancement to the official PrimeFaces open-source library**. (PR #12865)

SHMS (Smart Hotel Management System)

Nationwide licensing and registration system for hotels and restaurants

- > Contributed to the design and implementation of a **state-driven approval workflow**, reusable across multiple application types, minimizing code duplication
- > Utilized **AOP** to cleanly separate cross-cutting concerns like notifications from core business logic.

BCIC-ERP

End-to-end Integrated ERP system for process automation across 12 independent modules

- > **Assumed full ownership** of the Procurement and Inventory module, leading the process from business analysis (As-Is/To-Be modeling) and database schema design to hands-on development and successful client delivery.

Java JEE JSF Primefaces CDI EJB JPA PostgreSQL JMS Glassfish Prometheus Grafana Loki

★ STANDARDIZED TEST SCORES

TOEFL : 104/120 (Reading : 29, Listening : 27, Speaking : 23, Writing : 25)

PROJECTS AND OPEN SOURCE CONTRIBUTION

KNOWLEDGERELAY : AI-BASED KNOWLEDGE TRANSFER AGENT

2025

 github.com/riyadomf/knowledgeRelay

Built a RAG system to preserve project-specific knowledge base and assist new team members via contextual Q&A.

- > Engineered a multi-stage knowledge ingestion pipeline that automatically generates question-answer pairs from documents to fine-tune the retrieval process.
- > Integrated ChromaDB with vector embeddings and metadata-based filtering for source grounded efficient retrieval.
- > Implemented a contextualize_qa_chain to rephrase user queries using chat history for more accurate retrieval.

Python FastAPI LangChain ChromaDB OpenAI Ollama React

PRIMEFACES : JPALAZYDATAMODEL FEATURE ENHANCEMENT

2024

 github.com/primefaces/primefaces/pull/12865

Contributed to the PrimeFaces open-source library by adding the ability to inject custom FilterMeta into the JPALazyDataModel component. This enhancement allows developers to perform advanced, customizable filtering within DataTables.

Java JSF Primefaces JPA

SHOPAHOLIC

2022

 github.com/Nowshadjunaed/Shopaholic

Built a full-stack e-commerce platform connecting customers, banks, and suppliers.

JavaScript Node.js Express.js MongoDB React Redux

BREVITY

2021

 github.com/riyadomf/Brevity

A platform designed to connect beginners with experts in various fields to share knowledge, resources, and help others.

Python Flask MySQL JavaScript HTML CSS

TEACHING EXPERIENCE

Present
2024

Course Instructor, DYNAMIC LEARNING,

- > Authored and instructed a course - Enterprise Web Development with JEE, JSF, and PrimeFaces. 
- > Demonstrated core architectural patterns including component-driven design, CDI/EJB integration, backing bean scopes, and JSF lifecycle and state management.

RESEARCH EXPERIENCE

Shared Tasks

2025

Code Generation in Bangla – BLP Workshop, IJCNLP-AAACL 2025

- > **Low-Resource Adaptation** : Investigated low-resource language code generation by fine-tuning open-source models (Llama 3, TigerLLM, Qwen 2.5 Coder, Qwen 3) using parameter-efficient LoRA adapters.
- > **Reasoning Enhancement** : Enhanced model reasoning via Chain-of-Thought (CoT) augmented fine-tuning and a self-refinement loop that iteratively critiques and corrects generated code based on execution feedback.
- > **Error Analysis** : Conducted rigorous error analysis to identify and mitigate failure modes like instruction misinterpretation and overfitting, improving generalization and achieving 85% accuracy.

2023

Violence Inciting Text Detection (VITD) – BLP Workshop at EMNLP 2023

- > **Semi-Supervised Learning** : Applied self-training techniques to mitigate data scarcity and class imbalance, utilizing external datasets to improve model robustness.
- > **Data Augmentation** : Enhanced dataset diversity via back-translation, improving semantic variety and correcting linguistic inconsistencies.
- > **Ensemble Strategy** : Implemented a bagging and majority voting ensemble to reduce prediction variance and increase model stability.

PUBLICATIONS

Deja Vu at SemEval 2024 Task 9 : A Comparative Study of Advanced Language Models for Commonsense Reasoning

Accepted in *The SemEval 2024, NAACL, Mexico*

Team_Syrax at BLP-2023 Task 1 : Data Augmentation and Ensemble Based Approach for Violence Inciting Text Detection in Bangla

Accepted and Presented at *The BLP 2023, EMNLP, Singapore*

PROBLEM SOLVING

Solved over 500+ problems with C/C++ in different online judges

- > Vjudge:  [riyad2017331042](#) | LeetCode:  [riyad42](#) | LightOJ:  [riyadomf](#)
- > StopStalk Profile:  [riyadomf](#)

CERTIFICATIONS & ACHIEVEMENTS

- > 4th Place out of 32 teams – Code Generation in Bangla, A shared task at the BLP Workshop, IJCNLP-AACL 2025
- > 6th Place out of 30 teams – DSI AI Agent Hackathon, 2025
- > ISC2 Certified in Cybersecurity (CC)
- > Top 20 (Rank 19 → 12 post-eval) – Violence Inciting Text Detection (VITD), BLP 2023

EXTRA-CURRICULARS

- > Volunteer, IEEE International Conference on Bangla Speech and Language Processing (ICBSLP), 2019.
- > Ranked 11th in BUET CSE Fest 2022 AI Contest and 19th in Battle of Brains by Green University
- > Runners-Up, Intra-Department Football Tournament, 2022.